Matheus Augusto Tregnago

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Education

- **Master's Degree in Computer Science** — Mar. 2026 – Present
  Federal University of Rio Grande do Sul (UFRGS) — *Porto Alegre, Brazil*
  - Research Area: High-Performance Computing (HPC)
  - Advisor: Prof. Lucas Mello Schnorr
- **Bachelor's Degree in Computer Science** — Mar. 2022 – Dec. 2025
  University of Caxias do Sul (UCS) — *Bento Gonçalves, Brazil*
  - *Cum Laude* – GPA: 3.7/4.0

Professional Experience

- **Monsta Tecnologia** — Bento Gonçalves, RS, Brazil
  Full-Stack Developer — *Apr. 2025 – Present*
  - **Subscription and Payment Platform**: Redesigned the store, checkout, and subscription flows to support recurring products, multiple subscriptions per customer, and new products and services integrated with the Stripe API. Refactored the checkout state machine and redesigned the user interface to support more complex billing scenarios.
  - **Customer Journey Automation**: Designed and implemented an email automation system based on the customer lifecycle, with configurable rules based on subscription status, customer type, days until expiration, and time spent in each state, as well as customizable email templates managed through the administration panel.
  - **Dashboards and Monitoring**: Developed dashboards, charts, and interactive interfaces for real-time monitoring and visualization of devices and services.
  - **Interactive Device Map**: Developed an interactive and customizable map for visualizing the hierarchy of registered devices, with support for drag-and-drop, monitoring widgets, and configurable visual elements such as fonts, colors, and sizes.
- **Monsta Tecnologia** — Bento Gonçalves, RS, Brazil
  Technical Support Analyst — *Jul. 2022 – Apr. 2025*
  - **Technical Support**: Provided specialized remote support for the Monsta software platform, including system configuration, installation, and maintenance on Linux servers.
  - **Infrastructure**: Configured, maintained, and administered Linux servers, including service management, network ports, and firewall rules using firewalld and iptables.
  - **Troubleshooting**: Analyzed and interpreted system logs to diagnose failures, troubleshoot issues, and resolve incidents in production environments.
  - **Networking**: Analyzed and traced network traffic using tools such as tcpdump and netstat.
  - **Automation and Monitoring**: Developed and maintained Lua monitoring scripts integrated into the Monsta platform for tracking customer services and infrastructure.

Academic Publications

- **Parallel GPU Implementation for Coral Fractal Generation**: Matheus Tregnago, Samuel Francisco Ferrigo, and André Luís Martinotto. *Proceedings of the XXVI Regional School of High-Performance Computing of the Southern Region (ERAD-RS)*, 2026. Available at: https://sol.sbc.org.br/index.php/eradrs/article/view/42686
- **Evaluation of Performance Gains Using Parallelism on IoT Data from Smart Cities**: Matheus Tregnago, Ricardo Bregalda, and Samuel Francisco Ferrigo. *Proceedings of the Regional School of High-Performance Computing of the Southern Region (ERAD-RS)*, 2025. Available at: https://sol.sbc.org.br/index.php/eradrs/article/view/35354

Academic Activities

- **University of Caxias do Sul (UCS)** — Bento Gonçalves, RS, Brazil
  Calculus I Teaching Assistant — *Aug. 2023 – Jul. 2024*
  - **Academic Support**: Assisted students with topics including limits, derivatives, and integrals.
  - **Teaching**: Guided students through problem-solving, answered questions, and provided support with assignments and exam preparation.

Undergraduate Thesis

- **Parallel GPU Implementation for Coral Fractal Generation**
  Undergraduate Thesis – B.Sc. in Computer Science *2025*
  - **Description**: Developed a parallel GPU implementation for generating three-dimensional coral fractals using the Diffusion-Limited Aggregation (DLA) method.
  - **Technologies**: C++, CUDA
  - **Results**: Compared sequential and parallel implementations, achieving a speedup of up to 20×, with an analysis of performance limitations related to the maximum number of concurrently active GPU threads.

Projects

- **Mail2U Admin Interface**
  Professional Project – Email Management System *Aug. 2024 – Jan. 2025*
  - **Description**: Developed an administrative interface for the Mail2U email server, providing comprehensive management of domains, email accounts, and aliases.
  - **Backend**: Developed REST APIs with Express for creating, editing, and deleting accounts, aliases, and domains, including automatic password generation.
  - **Authentication and Authorization**: Implemented authentication using NextAuth, along with domain-level role-based access control and permissions.
  - **Database**: Designed the data model and implemented persistence using MariaDB.
  - **Status**: Production system currently used to manage real-world email accounts and domains.
  - **Technologies**: Next.js, Node.js, Express, MariaDB, NextAuth, shadcn/ui

Certifications

- **English Proficiency – B2 Level**: Michigan English Test (MET), 2021